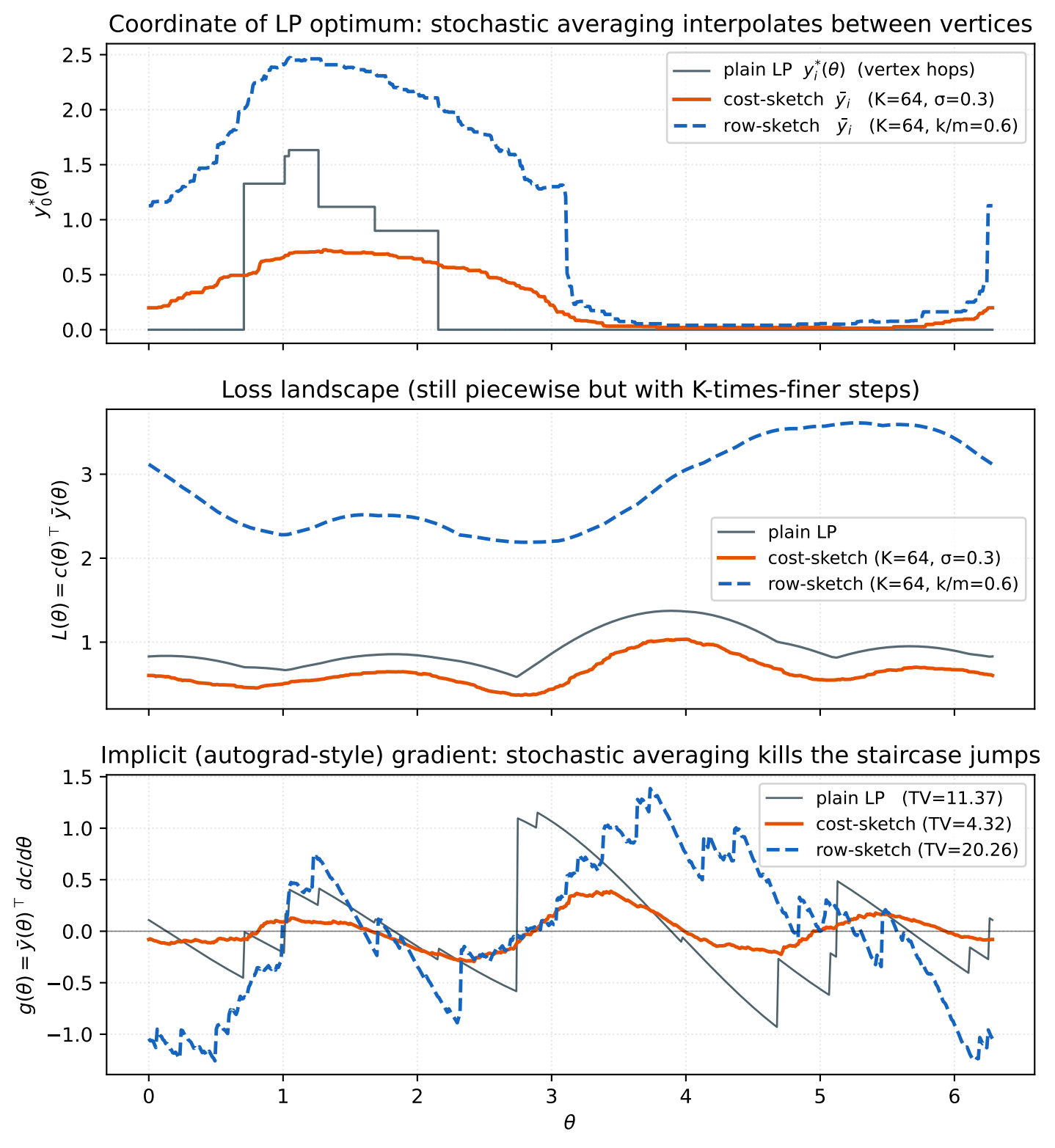
Coordinate of LP optimum: stochastic averaging interpolates between vertices
$y_0^*(\theta)$
plain LP $y_i^*(\theta)$ (vertex hops)
cost-sketch $\bar{y}_i$ (K=64, σ=0.3)
row-sketch $\bar{y}_i$ (K=64, k/m=0.6)
Loss landscape (still piecewise but with K-times-finer steps)
$L(\theta) = c(\theta)^\top \bar{y}(\theta)$
plain LP
cost-sketch (K=64, σ=0.3)
row-sketch (K=64, k/m=0.6)
Implicit (autograd-style) gradient: stochastic averaging kills the staircase jumps
$g(\theta) = \bar{y}(\theta)^\top dc/d\theta$
plain LP (TV=11.37)
cost-sketch (TV=4.32)
row-sketch (TV=20.26)
θ